

Roll No. :

MID SEMESTER EXAMINATION MARCH 2024

Name of the Program: **B.Tech**

Semester: 4th

Name of the Course:- Design and analysis of algorithms

Course Code: TCS 409

Time : 90 Minutes

Maximum Marks : 50

Note:

- i. Answer all the questions by choosing any one of the sub questions
- ii. Each questions carries 10 marks

Q1	(10 Marks)	CO1
(a)	What is asymptotic notations? Why do we use it to analyze time and space complexity instead of running time?	
	OR	
(b)	Analyze the following piece of code and find the space and time complexity. sum = 0; for(int i = 1; i <= n; i++){ for(int j = 1; j <= i*i; j++){ if(j%i == 0){ for(int k = 0; k < j; k++){ sum++; } } } }	
Q2	(10 Marks)	CO2
(a)	Write the pseudo code to find if a number is prime or not (most optimized approach). What is it's worst case time complexity? Arrange the following in ascending order of complexity $\lg(\sqrt{n!})$, n^2 , $\lg(n)$, $\sqrt{\lg(n)!}$, 2^n , 1.0001^n , $n*\lg(n)$	
	OR	
(b)	Write pseudo code for counting sort and explain why we cannot use it every time?	
Q3	(10 Marks)	

(a)	Write Tower of Hanoi pseudo code. Find the recurrence relation and determine the time and space complexity of the code.	CO2
	OR	
(b)	Write the pseudo code of insertion sort. What will be the output after 5 iterations of selection sort on the following numbers? 5,7,1,2,3,9,0,4,6	
Q4	(10 Marks)	CO1
(a)	Write code for merge sort. Draw and show how merge sort will work on the following array. 5,0,1,2,4,6,3,9	CO1
	OR	
(b)	Write the recursive code for binary search. Using recurrence relation find the time complexity of the code.	
Q5	(10 Marks)	CO1
(a)	Solve the following recurrence relations using backward substitution. Base case for all is $T(1) = 1$. 1. $T(n) = 2T(n-1) + n$ 2. $T(n) = T(n/2) + n$ 3. $T(n) = 4T(n/4) + n^2$	CO2
	OR	
(b)	Why do we tend to improve average case time complexity rather than worst case. Write the recursive code for finding the n^{th} Fibonacci number.	

Roll No.

TCS-408

B. TECH. (CSE) (FOURTH SEMESTER)
MID SEMESTER EXAMINATION, March, 2024

PROGRAMMING IN JAVA

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Illustrate why Java is called a “Portable” and “Robust” language.

Explain the components and execution process of a simple “hello world” program in Java. (CO2)

OR

- (b) Define classStudent with suitable data members, two constructors and member function. Create two objects using different constructors of the class. Also call member functions for each object and write the expected output of your program. (CO1)

P. T. O.

2. (a) Describe the following methods related to String/StringBuffer : (CO3)

(i) replace()

(ii) compareTo()

(iii) charAt()

(iv) indexOf()

(v) delete()

OR

(b) Write a Java program to implement a multilevel inheritance concept.

(CO5)

3. (a) Describe the uses of “final” and “super” keywords with respect to inheritance. (CO3)

OR

(b) Write a Java program to rearrange an array of strings in alphabetical order. (CO3)

4. (a) Compare Method Overriding and Method Overloading with the help of suitable examples. (CO3)

OR

(b) Write a Java program to create an abstract class BankAccount with abstract methods deposit() and withdraw(). Create subclasses : SavingsAccount and CurrentAccount that extend the BankAccount class and implement the respective methods to handle deposits and withdrawals for each account type. (CO5)

(3)

5. (a) What is an interface ? List the rules to create an interface in Java with an example program. (CO2)

OR

- (b) Define packages in Java. What are the advantages of using packages ?

Write syntax to create your own package in Java. (CO1)

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1. The first part of the report is a general introduction to the subject of the study.

2. The second part of the report is a detailed description of the methods used in the study.

OR

3. The third part of the report is a discussion of the results of the study.

4. The fourth part of the report is a conclusion and a list of references.

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1973

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TCS-402

B. TECH. (CSE) (FOURTH SEMESTER)
MID SEMESTER EXAMINATION, March, 2024

FINITE AUTOMATA AND FORMAL LANGUAGES

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

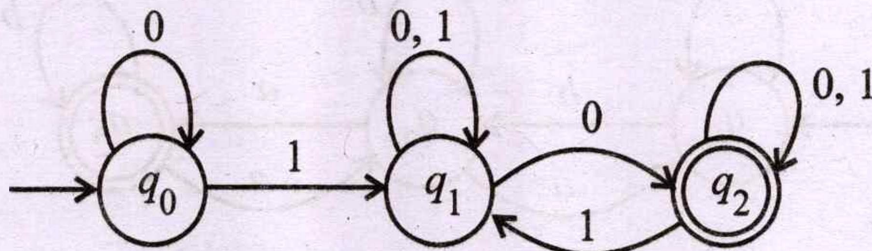
(ii) Each question carries 10 marks.

1. (a) Design a DFA over the alphabet $\Sigma = \{a, b\}$ such that it accepts all the strings where the number of a 's are divisible by 2 AND number of b 's are divisible by 3. (CO1)

OR

- (b) State and prove Kleene's Theorem for showing the equivalence between Regular Expression and Finite Automata. (CO1)

2. (a) Define NFA. Convert the given NFA to equivalent DFA : (CO1)

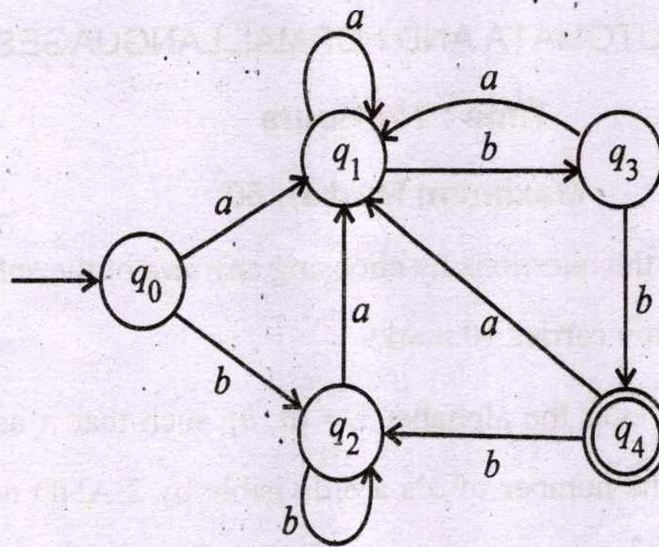


P. T. O.

OR

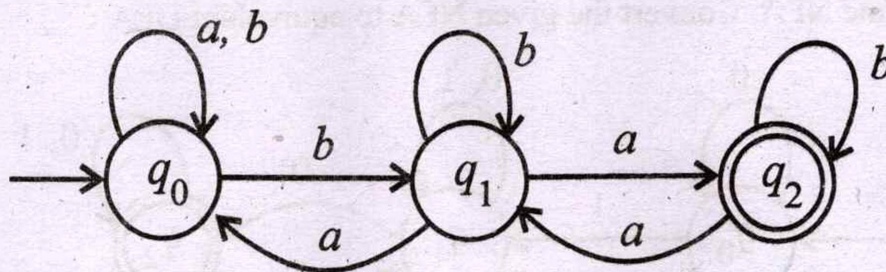
- (b) Design a DFA over the alphabet $\Sigma = \{0, 1\}$ such that it accepts all the binary strings where the equivalent decimal number is divisible by 8. (CO1)

3. (a) Write the steps in Myhill-Nerode theorem for minimization of DFA. Minimize the given DFA : (CO1)



OR

- (b) Find the Regular Expression for the given Finite Automata using Arden's Theorem : (CO1)



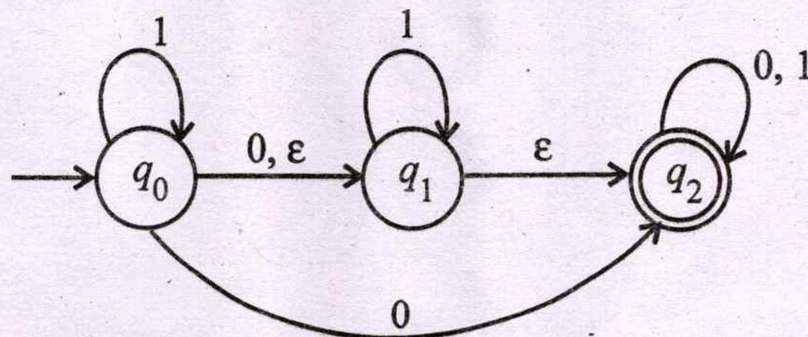
(3)

4. (a) Convert the given Mealy machine to equivalent Moore machine. Also, construct the state transition diagram for the Moore machine. (CO2)

Present State	Next State			
	Input $a = 0$		Input $a = 1$	
	Next State	Output	Next State	Output
$\rightarrow q_0$	q_1	0	q_2	0
q_1	q_1	0	q_2	1
q_2	q_1	1	q_2	0

OR

- (b) Construct a DFA over the alphabet $\Sigma = \{a, b\}$ such that it accepts all the strings of a's and b's that do not end with 'baba'. Test your machine for possible inputs. (CO2)
5. (a) Define epsilon-NFA. Convert the given epsilon-NFA into equivalent DFA : (CO2)



OR

- (b) State Pumping Lemma for Regular Languages. Prove that Language $\{L = 0^n 1^m 2^n, \text{ where } m \text{ and } n \text{ are arbitrary integers}\}$ is not regular.

(CO2)

Roll No.

TCS-403

**B. TECH. (CSE) (FOURTH SEMESTER)
MID SEMESTER EXAMINATION, March, 2024**

MICROPROCESSORS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) What is microcomputer ? How does the microprocessor differentiate between data and instruction ? (CO1 & CO2)

OR

- (b) Explain the functions of SID, HOLD, TRAP and IO/M signals of the 8085 microprocessor. (CO1 & CO2)

2. (a) Draw the timing diagram for instruction MOV M, D. (CO2 & CO5)

OR

- (b) Interface 4 kB EPROM with 8085 microprocessor using decoder. If output of active low 3 to 8 decoder is O6, then what will be the addressing range ? (CO2 & CO5)

3. (a) What will be value of ACC and status of flag register after execution of the following program ? (CO3)

MVI B 67
ADI B8
ANI 49
CMA

P. T. O.

(2)

OR

(b) Explain the difference between the following : (CO3)

(i) JC 2050 and CC 2050

(ii) RRC and RAR

4. (a) Write down any *five* instructions in 8085 which are related to 16-bit operation. Also determine addressing mode of each instruction. (CO3)

OR

(b) Write a program in 8085 which adds sum of two decimal numbers stored in memory location 2050 and 2051. If result is even, store the result at memory location 3050H else store the result at PORT 45.

(CO3)

5. (a) Explain the following code and determine the content of stack pointer and status of each flag after the execution of the following program :

(CO3)

```
LXI SP 209FH
MVI A F2H
XRI 47H
MOV C, A
PUSH B
POP PSW
HLT
```

OR

(b) Write an ALP using 8085 instructions to determine the sum of series of odd numbers in an array of 10 numbers. Store the result in a memory location starting at 3050H. Assume that numbers are stored from starting memory location 2050H. (CO3)

Roll No.

TCS-421

B. TECH. (CSE) (FOURTH SEMESTER)
MID SEMESTER EXAMINATION, March, 2024

FUNDAMENTAL OF STATISTICS AND AI

Time : 1½ Hours

Maximum Marks : 50

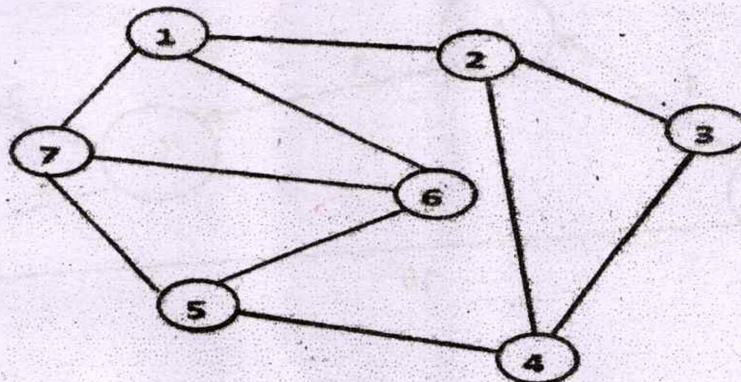
Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Define Data, Information, Knowledge as part of AI pyramid and give *two* use cases illustrating these steps. (CO2)

OR

- (b) Compare Uninformed search method and Informed search method. Find the path to reach the goal node (4) from the start node (1) by applying DFS : (CO3)



P. T. O.

2. (a) Describe and explain the types of AI in detail.

(CO1)

OR

- (b) Describe Intelligent Agent. Give PEAS description.

(CO2)

3. (a) Summarize the role of Artificial Intelligence with any *five* smart applications.

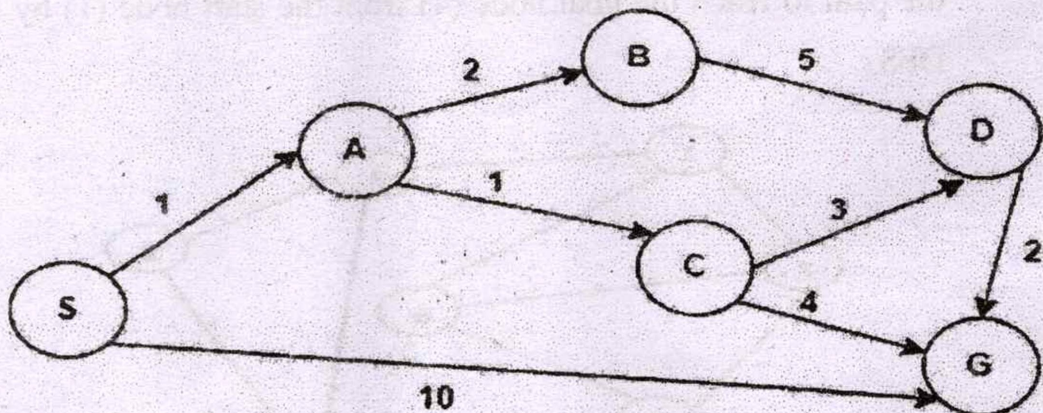
(CO2)

OR

- (b) Discuss A* search algorithm and next, apply the same for the given graph to search the goal node. Consider Start Node=S and Goal Node=G. Table below gives heuristic values :

(CO3)

S	5
A	3
B	4
C	2
D	6
G	0



(3)

4. (a) Elaborate the components of Turing tests for Artificial Intelligence and give *two* use cases. (CO2)

OR

- (b) Discuss hill climbing as problem solving in AI. (CO1)

5. (a) Define and explain AI. Explain problem solving and its role in AI.

(CO1)

OR

- (b) Discuss Iterative deepening depth-first search in detail with suitable examples. (CO2)

(1)

(a) Describe the requirements of the law for Artificial Intelligence and
(CO2)

(2)

(b) Describe the impact of the law on the development of
(CO1)

(c) Define and explain the concept of Artificial Intelligence and its role in AI
(CO1)

(3)

(d) Discuss the ethical implications of the law in detail with respect
(CO2)

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Roll No.

TCS-451

B. TECH. (CSE) (FOURTH SEMESTER)
MID SEMESTER EXAMINATION, March, 2024

VIRTUALIZATION AND CLOUD COMPUTING

Time : 1½ Hours

Maximum Marks : 50

- Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Founded in 2008, Nearmap is a location-intelligence company that uses aerial camera technology to help organizations in Australia, New Zealand, The United States and Canada track changes in their environments. Evaluate *four* main types of cloud computing in above mentioned organization. (CO1)

OR

- (b) Navan is a travel and expense software company that simplifies booking, viewing and managing business travel and expenses for thousands of companies around the world. Evaluate Cloud Computing Taxonomy with suitable diagram in above mentioned organization.

(CO1)

P. T. O.

2. (a) Flywire is a global payments enablement and software company that combines its proprietary global payments network, next-gen payments platform and vertical-specific software to deliver the most important and complex payments to global clients. Evaluate Cloud Computing Challenges in above mentioned organization. (CO3)

OR

- (b) Prime Video is a subscription-based, on-demand over-the-top streaming service, offered as part of a Prime subscription or as a stand-alone service. Launched in 2006, it has over 200 million customers. Evaluate Nine Steps Evolution of Cloud Computing in above mentioned organization. (CO2)

3. (a) Mediastream is a company that helps media and large companies to use streaming technologies easily. The company has been in the market for more than 15 years, with customers throughout Latin America, part of Europe and the U.S. Evaluate benefits of Virtualization Techniques in above mentioned organization. (CO3)

OR

- (b) Gilead Sciences Inc. (Gilead) modernized its data infrastructure and improved its operational performance through a data mesh framework built on AWS, reducing data search times by over 50 percent. Evaluate six types of Virtualization Techniques in above mentioned organization.

(CO3)

(3)

4. (a) Neeva, an ad-free search engine, uses Karpenter and Amazon EC2 Spot Instances to maintain cost optimization and reduce the time spent waiting on infrastructure management by 10–100 hours per week. Evaluate full virtualization with suitable diagram in above mentioned organization. (CO2)

OR

- (b) Merck KGaA, Darmstadt, Germany improves its time to data insights with a connected data and analytics platform on AWS, helping data teams access secure, custom environments faster. Evaluate para-virtualization with suitable diagram in above mentioned organization. (CO3)

5. (a) Mircom, a global designer, manufacturer and distributor of intelligent building solutions, wanted to modernize its Open Graphic Navigator (OpenGN)—a single-site digital twin and on-premises Internet of Things (IoT) software platform. Evaluate Microsoft Hyper-V with suitable diagram in above mentioned organization. (CO3)

OR

- (b) Fatshark, a Swedish video game developer, wanted to build its most complex game yet—Warhammer 40,000 : Darktide. To build on the success of the studio's Warhammer : Vermintide series, the combat-focused cooperative multiplayer game must offer ultralow latency to over 1,00,000 concurrent players. Evaluate VMware with suitable diagram in above mentioned organization. (CO1)

1. The first step in the process of the investigation is the selection of the subject. This is done by the investigator who is assigned to the case. The subject is then interviewed and the information obtained is used to develop a plan of action. The plan of action is then carried out and the results are reported to the investigator.

2. The second step in the process of the investigation is the collection of evidence. This is done by the investigator who is assigned to the case. The evidence is then analyzed and the results are reported to the investigator.

3. The third step in the process of the investigation is the identification of the suspect. This is done by the investigator who is assigned to the case. The suspect is then interviewed and the information obtained is used to develop a plan of action. The plan of action is then carried out and the results are reported to the investigator.

4. The fourth step in the process of the investigation is the arrest of the suspect. This is done by the investigator who is assigned to the case. The suspect is then interviewed and the information obtained is used to develop a plan of action. The plan of action is then carried out and the results are reported to the investigator.

Roll No.

TCS-471

B. TECH. (CSE/B. TECH.) (BIO-TECH.)

(FOURTH SEMESTER)

MID SEMESTER EXAMINATION, March, 2024

STATISTICAL DATA ANALYSIS WITH R

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) A number of particular articles has been classified according to their weights. After drying for two weeks the same articles have again been weighted and similarly classified. It is known that the median weight in the first weighing was 20.83 kilogram while in the second weighing it was 17.35 kilogram. Some frequencies a and b in the first weighing and x and y in the second are missing. It is known that $a = x/3$ and $b = y/2$. Find out the values of the missing frequencies. (CO1)

P. T. O.

Class	Frequencies	
	1st weighing	2nd weighing
0—5	a	x
5—10	b	y
10—15	11	40
15—20	52	50
20—25	75	30
25—30	22	28

OR

- (b) A special dice is prepared such that the probabilities of throwing 1, 2, 3, 4, 5 and 6 points are : (CO1)

$$\frac{1-k}{6}, \frac{1+2k}{6}, \frac{1-k}{6}, \frac{1+k}{6}, \frac{1-2k}{6} \text{ and } \frac{1+k}{6}$$

respectively. If two such dice are thrown, find the probability of getting a sum equal to 9.

2. (a) In a bolt factory machines A, B and C manufacture respectively 25%, 35% and 40% of the total. of their output 5, 4, 2 percent are defective bolts. A bolt is drawn at random from the product and is found to be defective. What are the probabilities that it was manufactured by machines A, B and C ? (CO1)

(3)

TCS-471

OR

(b) Explain the following :

(CO1)

(i) Statistics as 'Statistical Data'

(ii) Statistics as 'Statistical Methods'

(iii) Importance and scope of Statistics

(iv) Limitations of Statistics

(v) Distrust of Statistics

3. (a) p is the probability that man aged x years will die in a year. Find the probability that out of n men $A_1, A_2, \dots, A_n$ each aged x , A_1 will die in a year and will be the first to die.

(CO1)

OR

- (b) Explain the methods of manipulation and data processing in R programming language with a numerical example.

(CO2)

4. (a) Explain the methods of reading and getting data in R programming language with an example.

(CO2)

OR

- (b) Explain the various types of data types in programming language.

(CO2)

P. T. O.

5. (a) Compare standard packages and command packages in R programming language. Elaborate the steps for installing extra packages for windows users. (CO2)

OR

- (b) Explain the methods of exporting data in programming language R with a possible numerical example. (CO2)

Roll No.

TCS-492

B. TECH. (CSE) (FOURTH SEMESTER)
MID SEMESTER EXAMINATION, March, 2024
FUNDAMENTALS OF CYBER SECURITY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) What measures can be implemented to ensure the confidentiality of sensitive data in transit over a network ? How can Distributed Denial of Service (DDoS) attacks impact the availability of online services and what mitigation strategies can be employed ? (CO1)

OR

- (b) Explain the concept of the Zero Trust Model and its key principles. (CO1)

2. (a) Discuss the types of hackers in the context of Cyber Security. Provide examples of their activities. (CO1)

OR

- (b) What is the main objective of the 'gaining access' phase in penetration testing and what methods are commonly employed to achieve it ? Why might maintaining access be a relevant objective in a penetration test and what challenges does it pose for the tester ? (CO1)

P. T. O.

(2)

3. (a) Define and differentiate between lists and tuples in Python. Provide examples of situations where each data structure is more suitable. (CO2)

OR

- (b) What measures can be implemented to ensure the integrity of critical system files and configurations against 'unauthorized modifications' ?

(CO2)

4. (a) Explain the process of adding and deleting users in a Linux system. Discuss the importance of adding users to the 'sudo' group for security purposes. (CO2)

OR

- (b) Explain the process of creating, copying, moving and removing files in Linux. How can these file operations be utilized in ethical hacking scenarios ? (CO2)

5. (a) What strategies can be employed to create backups that safeguard against accidental or intentional file deletion ? (CO1)

QR

- (b) How can the principle of 'least privilege' be applied to limit user access and reduce the risk of unauthorized entry into sensitive areas ? (CO1)